Day 3: RAM and Storage Theory (30 min): Types of RAM (DDR3, DDR4, DDR5) and storage (HDD vs. SSD, NVMe).Practical (30 min): Remove and reinstall RAM and storage drives.

1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

ANS:

Feature	DDR3	DDR4	DDR5
Speed	800–2133 MT/s	1600–3200 MT/s	4800–8800+ MT/s
Voltage	1.5 V	1.2 V	1.1 V
Performance	Lower	Better	Highest of the three
Power use	Higher	Lower	Lowest standard voltage
Typical use	Older computers	Many desktop PCs and laptops	Newer computers and gaming PCs
Availability	Older generation	Widely used	Newer generation

**2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.**



STEP 1: RAM Installed

STEP 2: RAM Removed (Release the clips)

STEP 3: RAM Removed

STEP 4: RAM Reinstalled

STEP 5: System Started

3.List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

ANS:

Feature	HDD	SSD (SATA)	NVMe SSD
1. Speed	Slow	Fast	Very fast
2. Price	Cheapest	More expensive than HDD	Usually more expensive than HDD
3. Technology	Uses spinning disks	Uses flash memory	Uses flash memory with a faster connection
Daily use case	Storing photos and videos	Opening apps like Spotify	Gaming or opening large apps quickly

**4.Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model. .**

ANS:



STEP 1: Locate the HDD/SSD

STEP 2: Identify the SATA connector

STEP 3: Storage drive removed

STEP 4: Storage drive removed

STEP 5: Check storage detection